PHISHING EMAIL DETECTION & AWARENESS REPORT

Task Two

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FUTURE INTERNS

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1. Introduction

Phishing is one of the most common cyber threats affecting individuals and organizations today. Attackers use deceptive emails to trick users into revealing sensitive information such as passwords, financial data, or personal details.

These attacks are highly effective because they exploit human behavior rather than technical vulnerabilities. As a result, organizations invest in phishing detection and employee awareness programs to reduce risk.

This report analyzes real-world phishing email examples obtained from public cybersecurity awareness sources and used for educational purposes. The emails are examined to identify common phishing indicators, classify their risk level, and explain how these attacks work in a clear and simple manner.

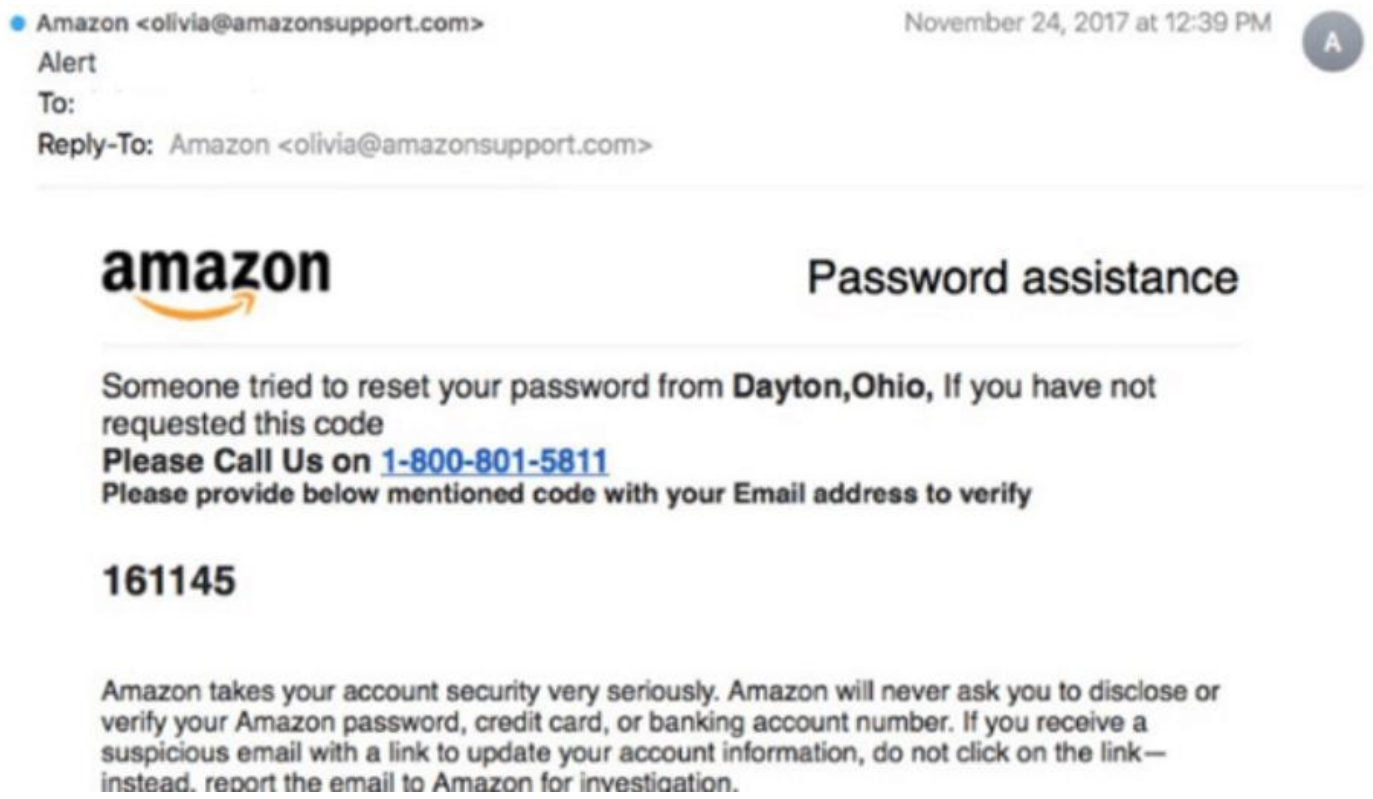
The analysis also demonstrates how basic investigation techniques, such as email content review, header inspection, and link analysis, can be used to detect phishing attempts. Based on these findings, practical prevention guidelines are provided to help users recognize and avoid phishing attacks.

2. Tools Used:

Category:	**Tool:**
1. Email Header Analysis Tools:	• Google Message Header Analyzer Used to analyze email headers and identify authentication results such as SPF, DKIM, and DMARC. • MXToolbox Email Header Analyzer Used to perform detailed header analysis, including email routing, blacklist checks, and authentication validation.
2.URL and Link Analysis Tools:	• VirusTotal Used to scan suspicious URLs and determine whether they are flagged as malicious or associated with phishing activity. • Browser Inspection Techniques Used to safely examine links without directly interacting with potentially harmful websites.
3. Data Sources:	• Public phishing email samples obtained from cybersecurity awareness repositories (used strictly for educational purposes). • Sample datasets containing real-world phishing email examples.
4.Documentation Tools:	• Microsoft Word: Used to create and format the report. • GitHub: Used to store and submit the final project and supporting evidence.

3. Phishing Email Analysis:

3.1. Example 1: Amazon



Amazon <olivia@amazonsupport.com> November 24, 2017 at 12:39 PM

Alert

To:

Reply-To: Amazon <olivia@amazonsupport.com>

amazon

Password assistance

Someone tried to reset your password from **Dayton,Ohio,** If you have not requested this code

Please Call Us on 1-800-801-5811

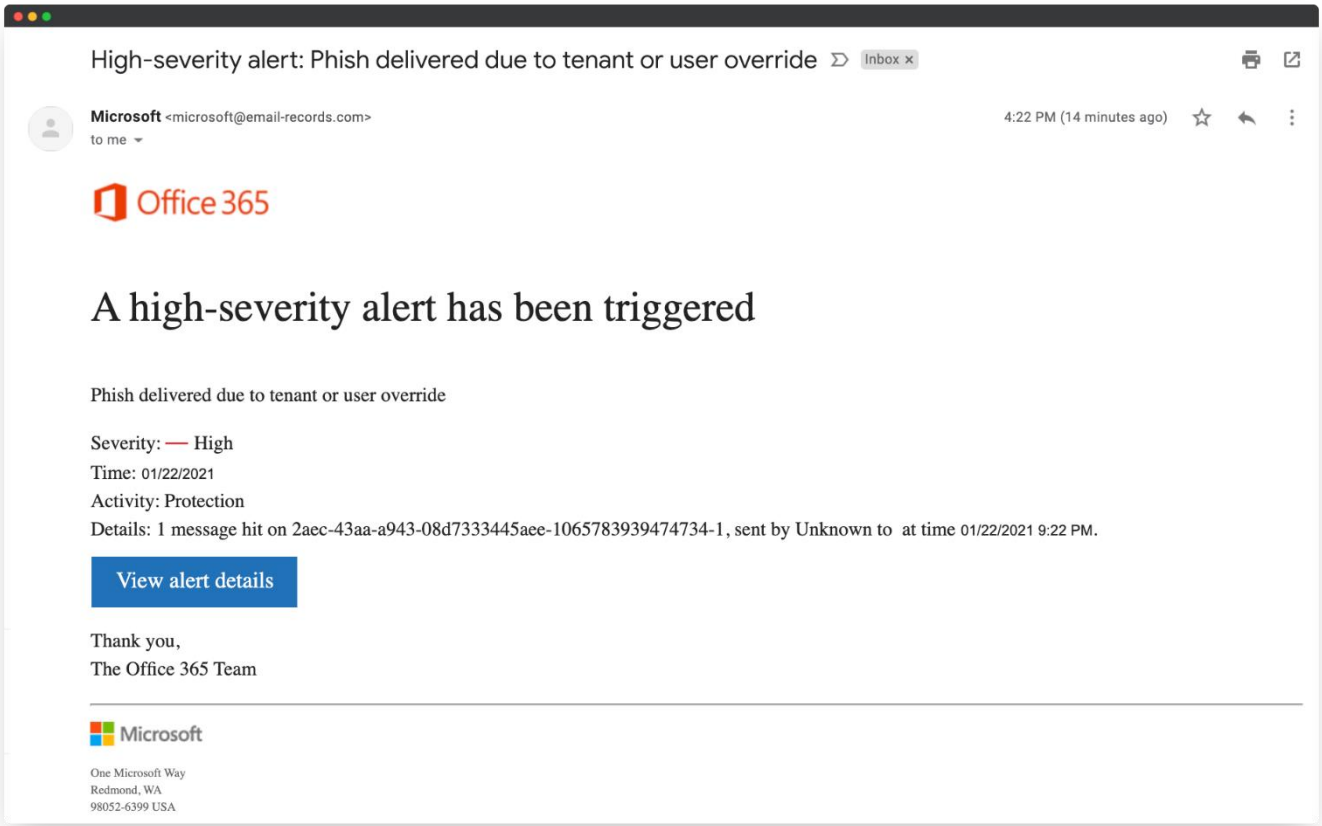
Please provide below mentioned code with your Email address to verify

161145

Amazon takes your account security very seriously. Amazon will never ask you to disclose or verify your Amazon password, credit card, or banking account number. If you receive a suspicious email with a link to update your account information, do not click on the link— instead, report the email to Amazon for investigation.

Phishing Indicators:	• Spoofed sender pretending to be Amazon • Suspicious link not matching official domain • Urgent message regarding account issues • Generic greeting
Risk Classification:	Phishing
Explanation:	This email impersonates Amazon to trick users into clicking a malicious link. The attacker creates urgency to pressure the user into acting quickly. The link likely leads to a fake login page designed to steal credentials.

3.2. Example 2: Office 365



High-severity alert: Phish delivered due to tenant or user override Inbox ×

Microsoft <microsoft@email-records.com>
to me

4:22 PM (14 minutes ago)

Office 365

A high-severity alert has been triggered

Phish delivered due to tenant or user override

Severity: — High
Time: 01/22/2021
Activity: Protection
Details: 1 message hit on 2aec-43aa-a943-08d7333445aee-1065783939474734-1, sent by Unknown to at time 01/22/2021 9:22 PM.

View alert details

Thank you,
The Office 365 Team

Microsoft

One Microsoft Way
Redmond, WA
98052-6399 USA

Phishing Indicators:	• Fake Microsoft branding • Request to log in urgently • Suspicious login URL • Generic greeting
Risk Classification:	Phishing
Explanation:	This email impersonates Microsoft Office 365 and attempts to steal login credentials. Users are redirected to a fake login page that captures their information.

3.3. Example 3: Suspicious Activity Alert

From: "MS-Support Centre" <outlook_2C5A4DD15A5A1106@outlook.com>
Sent: Tuesday, June 2, 2020 11:30:27 AM
Subject: Account unusual sign-in activity

Microsoft account

Suspicious Account Acitivty

Hello

This is to inform you that we have found suspicious activities with your account.Due to that, we have terminated your windows account.

Please find suspicious incident details:
Recent Incident Details: Eastern Belarus(IP Address : 10.97.87.25)
MAC Address 01:AD:99:00 & IP: 10.97.87.25

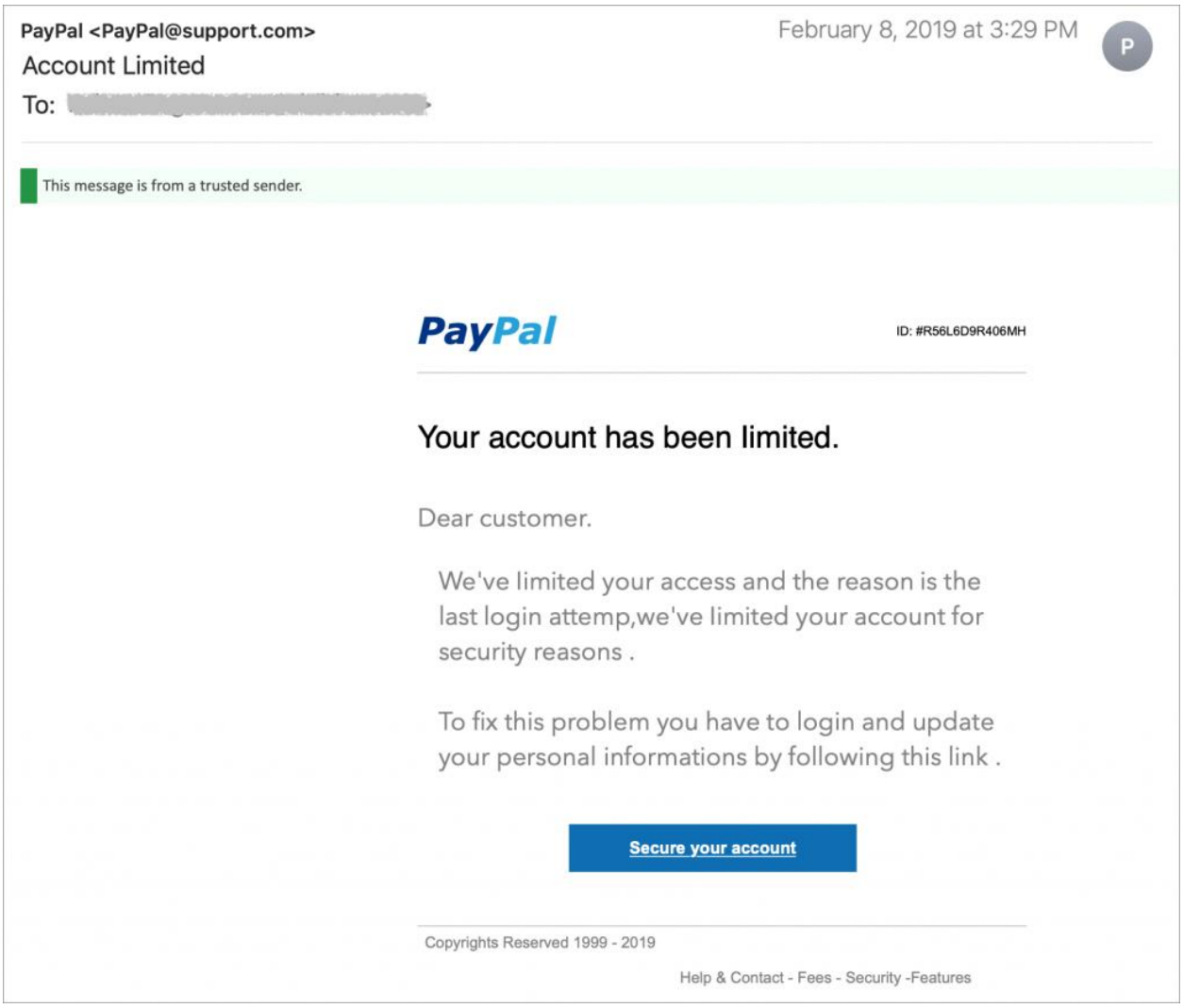
If you think this was a mistake and you wish to continue using this windows license key, Please contact con technical support at **1-800-341-8835.**

PS NOTE: Please be at your computer while you call consumer technical support.

Windows Help
1-800-341-8835.

Phishing Indicators:	• Fear-based message (“suspicious activity”) • Urgent request for verification • Suspicious link • Generic greeting
Risk Classification:	Phishing
Explanation:	This email uses fear tactics to trick users into clicking a malicious link. The goal is to steal sensitive account information.

3.4. Example 4: PayPal



PayPal <PayPal@support.com>

February 8, 2019 at 3:29 PM

Account Limited

To: [redacted]

This message is from a trusted sender.

PayPal

ID: #R56L6D9R406MH

Your account has been limited.

Dear customer.

We've limited your access and the reason is the last login attemp,we've limited your account for security reasons .

To fix this problem you have to login and update your personal informations by following this link .

Secure your account

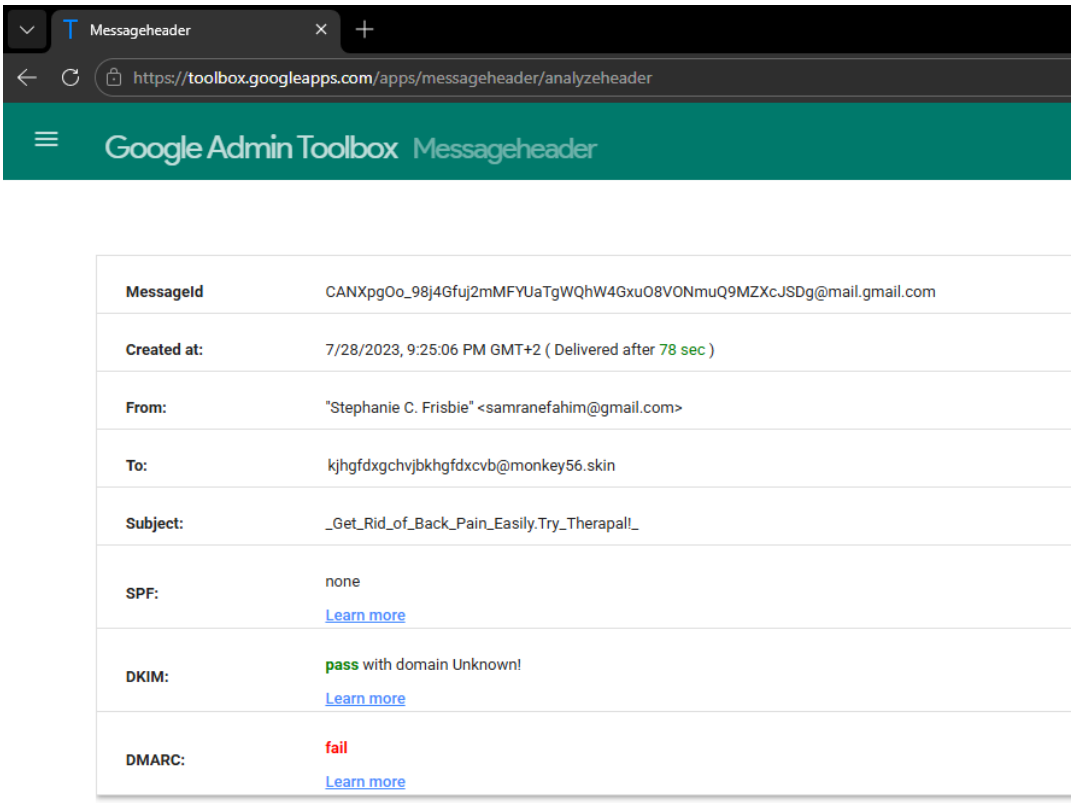
Copyrights Reserved 1999 - 2019

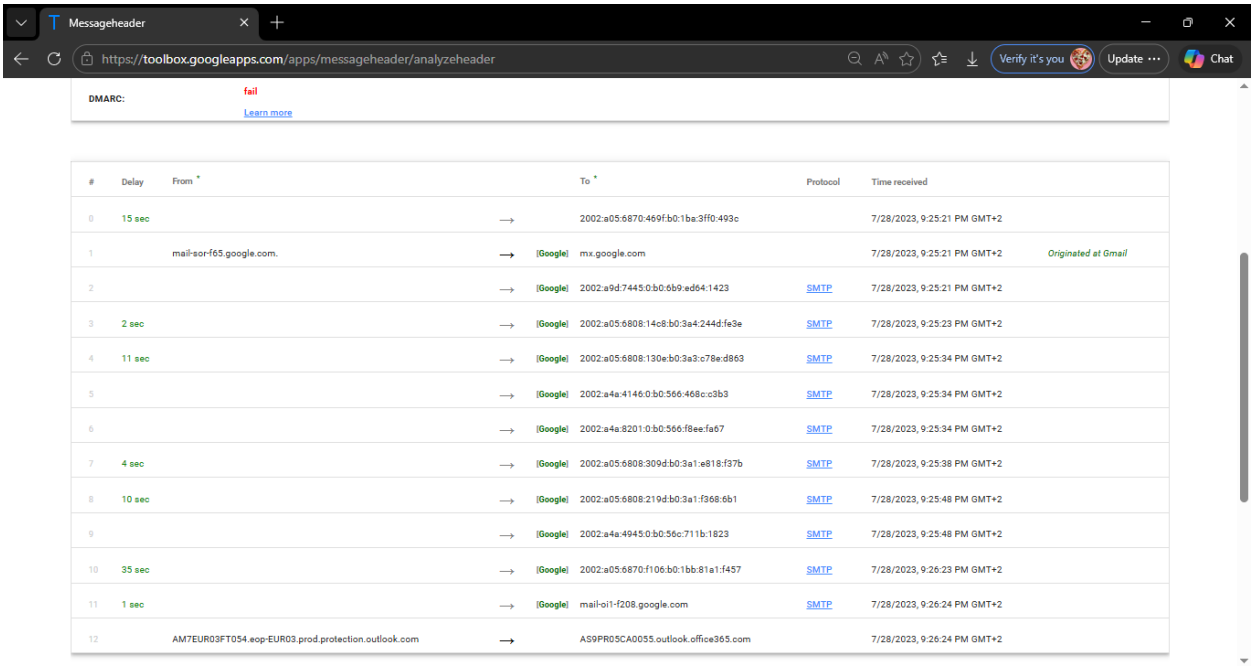
Help & Contact - Fees - Security -Features

Phishing Indicators:	• Poor grammar and formatting • Unknown sender • Suspicious request • Generic greeting
Risk Classification:	Phishing
Explanation:	This email demonstrates common phishing characteristics such as poor language and suspicious intent. Attackers rely on users ignoring these warning signs.

3.5. Example 5: Sample 1 (Sample – 1011.eml)

Source of Sample: GitHub Repository - Phishing Pot Dataset





Email Header Analysis: Sample – 1011.eml

Tool Used:	Google Message Header Analyzer
Key Findings from Header Analysis - Sample – 1011.eml	
1.Sender Identity Mismatch: (Spoofing)	•Display Name: Stephanie C. Frisbie •Email Address: samranefahim@gmail.com The display name does not match the actual email address, which is a common phishing technique used to mislead users.
2.SPF Result:	• Spf=none Domain 'monkey56.skin' is not authorized This means the sender is not verified or authorized to send emails on behalf of the domain.
3.DMARC Result:	• dmarc=fail This indicates that the email failed authentication and is likely to be fraudulent.
4. Suspicious Subject Line:	• Subject: "Get Rid of Back Pain Easily. Try Therapal!" The subject appears promotional and unrelated to a trusted organization, which is commonly seen in spam or phishing emails.
5. Suspicious Recipient Domain:	• 'monkey56.skin' The email was sent to a non-standard domain, which appears random and untrustworthy.
6. Use of Free Email Service:	The email was sent from a Gmail account rather than a professional or organizational domain. Legitimate businesses typically use official domain email addresses.
Risk Classification:	Phishing
Explanation:	This email shows multiple signs of a phishing attempt. The sender uses a fake display name and a free email account to appear trustworthy. Authentication checks failed, meaning the email cannot be verified as legitimate. These techniques are commonly used by attackers to trick users into trusting the email and interacting with malicious content.

Browser Tools (URL Analysis) - Sample – 1011.eml	
Tool Used:	VirusTotal
Links Analyzed:	1. https://tinyurl.com/2jcccaj3 2. https://tinyurl.com/mvejw6rf
Key Findings from URL Analysis - Sample – 1011.eml	
1.Use of URL Shortening Service:	Both links use **TinyURL**, which hides the real destination. This is a common phishing technique used to: • Conceal malicious websites • Prevent users from verifying links
2.Malicious Detection (Link 1):	• 2 out of 97 security vendors flagged the URL as malicious • Detected as: **Phishing** Even a small number of detections is important, as it indicates **known suspicious activity.**
3. Malicious Detection (Link 2):	• 1 out of 97 security vendor flagged the URL as malicious • Detected as: Phishing Indicates potential risk and association with phishing campaigns
4. Hidden Destination Risk:	Because the links are shortened, users cannot see: • The real website • Whether it is legitimate or fake This increases the likelihood of users being redirected to Fake login pages or malware downloads.
Risk Classification:	Phishing
Explanation:	The email contains shortened links that hide the actual destination. This makes it difficult for users to determine whether the links are safe. Security analysis shows that these links have been flagged as phishing by multiple security vendors. This indicates that the links may lead to malicious websites designed to steal sensitive information or infect devices. URL analysis revealed that the links were shortened and flagged by security vendors as phishing, confirming the presence of malicious intent.

3.6. Example 6: Sample 2 (Sample – 1004.eml)

Source of Sample: GitHub Repository - Phishing Pot Dataset

Email Header Analysis – Sample – 1004.eml	
Tool used:	Using MXToolbox Email Header Analyzer
Key Security Findings - Email Header Analysis – Sample – 1004.eml	
1. DMARC Failure:	• DMARC policy: p=quarantine • Result: **Fail** This is a **critical issue** as this means the email **failed authentication checks,** it should be treated as **untrusted and potentially malicious.**
2. SPF Failure:	• SPF result: SoftFail • Sending IP: 159.27.24.86 This IP is **Not listed** in the domain's allowed senders. Indicates **email spoofing as the sender is not authorized.**
3. No DKIM Signature:	• DKIM: None The email was **not digitally signed, t**here is **no way to verify integrity.**
4. Blacklisted Sending Servers:	From relay analysis: • 147.78.103.9 - Blacklisted • 159.27.24.86 - Blacklisted This is very serious as blacklisted IPs are commonly used in Spam campaigns, Phishing attacks, Malware distribution.
5. Suspicious Email Route:	Email originated from an unknown user system: **User (147.78.103.9).** It has been passed through a suspicious domain: **mail.mail04.zhanlingol.com** The email did not originate from a trusted corporate server and is likely sent from a compromised or malicious system.
6. Mismatched Identity:	• **From:** ghulammustafa@cyber.net.pk • **Reply-To:** philipffredrick3690@gmail.com The User thinks they are replying to a legitimate domain, but response goes to attacker-controlled Gmail.
7. Spam Confidence Level (SCL):	• SCL: 7 This indicates there is a high probability of spam or phishing.
Risk Classification:	Phishing - **High Risk**

	Reasons: • DMARC, SPF, DKIM failures • Blacklisted sender IP • Suspicious email routing • Mismatched sender identity • Fraudulent financial offer The MXToolbox analysis confirms multiple authentication failures and blacklisted sender infrastructure, strongly indicating that the email is part of a phishing campaign rather than legitimate communication.

4. Phishing Prevention and Awareness Guidelines:

Phishing attacks rely heavily on human error rather than technical vulnerabilities. Therefore, user awareness and proper security practices are essential in preventing successful attacks. The following guidelines provide clear actions users and employees should follow to reduce the risk of phishing.

4.1 Prevention Tips for Users:

Users should adopt the following best practices when dealing with emails:

- Always verify the sender's full email address, not just the display name.
- Be cautious of emails from unknown or suspicious domains.
- Avoid clicking on links in unsolicited or unexpected emails.
- Hover over links to preview the destination before clicking.
- Avoid interacting with shortened URLs (e.g: TinyURL).
- Do not download attachments from untrusted sources.
- Be cautious of urgent, threatening, or "too good to be true" messages.
- Use security tools (e.g: VirusTotal) to scan suspicious links.
- Access important websites (e.g: banking or email) directly through the browser instead of email links.
- Keep software, antivirus, and browsers updated.

4.2 Do’s and Don’ts for Employees:

Do’s:

- Verify email authenticity using SPF, DKIM, and DMARC indicators (if available).
- Report suspicious emails to the IT or security department immediately.
- Follow organizational security policies and procedures.
- Double-check requests involving sensitive information or financial transactions.
- Use multi-factor authentication (MFA) where possible.
- Attend cybersecurity awareness training regularly.

Don’ts:

- Do not click on suspicious links or attachments.
- Do not reply to unknown or untrusted senders.
- Do not share personal, login, or financial information via email.
- Do not trust emails offering money, prizes, or urgent requests.
- Do not bypass security warnings or ignore browser alerts.
- Do not use unsecured or public networks for sensitive activities.